

Meaning Theory:

Semantic Stability in Human Systems

A Synthesis of Cultural and Organizational Analysis

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Definition: Meaning Theory is the study of when semantic structure remains stable under amplification.

Abstract

This paper extends Meaning Theory into human systems, synthesizing findings from two complementary analyses: a comparative semantic examination of cultural and idiomatic expressions, and an organizational study of semantic drift in management and coordination structures.

Together, these analyses demonstrate that semantic stability failures in human systems follow patterns structurally consistent with those observed in the prior Meaning Theory pilot experiment on artificial systems. The same framework — $M = S \times A \times C$ — appears to predict semantic preservation and collapse across cultural transmission, cross-linguistic translation, and organizational amplification chains.

This paper does not present formal human-subject experiments. It performs operational analysis of documented phenomena to demonstrate consistency with the Meaning Theory framework and to extend its scope beyond artificial systems.

Background

The preceding Meaning Theory pilot experiment (ScrollBearer8, 2026) demonstrated semantic drift in artificial systems through recursive paraphrase amplification across five frontier language models.

The central finding was a clear gradient: structured texts with strong semantic anchoring preserved meaning under repeated amplification, while vague or weakly anchored texts drifted more rapidly. Drift behavior was consistent within each model but varied between models, suggesting that semantic drift pattern is a stable architectural property rather than random noise.

This paper asks the same question of human systems. Two parallel domains of inquiry are combined:

- Cultural and idiomatic expressions, where meaning depends on shared amplification systems (cultural context, community membership, historical transmission)
- Organizational and management structures, where meaning passes through human coordination chains and is subject to interpretive drift at each transmission node

Across both domains, the central question is identical to the one posed in the AI pilot:

Under what conditions does semantic structure remain stable under amplification?

Clarification on Meaning

Meaning Theory does not treat meaning as a single phenomenon.

Instead, meaning may be understood as a layered system:

- Structural Meaning — stability under amplification
- Epistemic Meaning — correspondence to reality
- Normative Meaning — value and evaluation
- Experiential Meaning — felt significance
- Existential Meaning — consequence under irreversibility

The present paper focuses exclusively on Structural Meaning.

Under this definition, meaning refers to the degree to which semantic structure remains stable as it passes through amplification systems such as artificial intelligence, cultural transmission, translation, organizational hierarchies, and human communication.

The framework does not attempt to measure subjective experience, moral value, truthfulness, or existential significance. These may represent higher layers of meaning that operate independently of structural stability.

This distinction allows Meaning Theory to analyze semantic preservation across both biological and artificial systems without making claims regarding consciousness, subjective experience, or existential purpose.

Framework

Meaning Theory proposes that semantic stability is determined by the interaction of three variables:

$$M = S \times A \times C$$

S — *Structural coherence of the message*

A — *Amplification: the process by which semantic content is propagated, reinforced, interpreted, transmitted, or operationalized across a system (cultural, organizational, artificial, or communicative).*

C — *Clarity of specification and constraint*

The multiplicative form predicts that sufficiently weakening any single variable should produce semantic collapse regardless of the strength of the remaining variables.

In artificial systems, amplification takes the form of recursive re-encoding through model paraphrase. In human systems, amplification emerges through:

- Cultural interpretation and community membership
- Idiomatic compression and metaphorical transmission
- Translation across linguistic systems
- Management hierarchies and delegation chains
- Organizational procedures, meetings, and documentation

The mechanism differs between domains. The underlying dynamic is proposed to be the same.

Part I — Cultural and Idiomatic Semantic Stability

1.1 Baseline: High Structural Stability

Ordinary factual statements exhibit high semantic stability across transmission because they rely on explicit structural content rather than shared amplification systems.

English: "The chocolate is on the table."

French: "Le chocolat est sur la table."

Semantic recovery is stable across translation because intended meaning is fully carried by the structure. Amplification (A) is not required to recover the meaning. This corresponds to the structured texts in the AI pilot, which preserved meaning across all five models under three rounds of recursive paraphrase.

1.2 Idiomatic Expressions: Amplification Dependency

Idiomatic expressions preserve literal grammatical structure while making intended meaning unrecoverable without the shared amplification system of the originating culture.

English: "Break a leg"

Literal: *Physically injure yourself.*

Intended: *Good luck.*

Meaning is recoverable within English-speaking theatrical culture. Outside that amplification system, semantic recovery fails despite grammatical comprehensibility. Under the framework: high S, high C, but low A outside originating context → semantic opacity.

French: "Chercher midi à quatorze heures"

Literal: *To look for noon at 2 PM.*

Intended: *To unnecessarily complicate something.*

Grammatical structure is fully transparent. Intended meaning is unrecoverable outside the cultural amplification system that created and sustained the expression.

Japanese: "空気を読む" (Kūki wo yomu)

Literal: *Read the air.*

Intended: *Understand unspoken social context.*

This expression encodes an entire social operating system — the expectation of implicit contextual awareness — that has no direct structural equivalent in many other languages. The amplification system here is not merely cultural but socially constitutive.

Observation: across all idiomatic cases, literal syntax survives intact while intended semantic recovery collapses without shared amplification. The pattern corresponds directly to what Meaning Theory predicts under low A conditions.

1.3 Unification of Prior Research Findings

Multiple research traditions have independently documented semantic instability under transmission across human systems. Meaning Theory proposes these may represent different failure signatures of the same underlying relationship.

Failure Condition	Predicted Outcome	Human Example	AI Analog
Low S (structure)	Fidelity decay	Vague delegation chains	Vague text paraphrase collapse
Low A (amplification)	Semantic opacity	Cross-cultural idiom failure	Context-window compression drift

Low C (clarity)	Schema-ward drift	Micromanagement intent reversal	Abstract text commitment drift
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Bartlett's serial reproduction studies (1932) showed schema-ward drift: unfamiliar narratives reorganized toward culturally familiar structures during repeated retelling — a low-C failure. Mesoudi and Whiten's cultural transmission studies demonstrated fidelity decay in structurally weak information — a low-S failure. Cross-cultural pragmatics documented semantic opacity when contextual reinforcement was absent — a low-A failure.

Under the Meaning Theory framework, these are not competing explanations but complementary observations of the same dynamic under differing amplification conditions.

Part II — Organizational Semantic Stability

2.1 Organizations as Amplification Systems

Human organizations are, among other things, semantic transmission systems. Instructions, intentions, and goals pass through chains of human interpretation before becoming operational behavior.

Each transmission node — a meeting, a report, a management decision, a delegation step — introduces an opportunity for semantic preservation or semantic drift. Unlike machine amplification, organizational amplification involves active interpretation by agents with their own goals, schemas, and contextual constraints.

This makes organizational systems a distinct but structurally analogous amplification domain.

2.2 Case Studies in Organizational Semantic Drift

Case 1 — Delegation Drift

A single executive instruction passes through four organizational levels:

Executive intent: *"Improve customer response times."*

- Director interpretation: "Reduce ticket backlog."
- Manager interpretation: "Increase ticket closure rates."
- Employee interpretation: "Close tickets as quickly as possible."

The original intent focused on customer outcomes. The recovered intent at the operational level focused on numerical throughput. Meaning drift occurred despite continuous communication through a functioning organizational structure.

Under the framework: high S (the instruction was clear), low C (criteria for success were not specified), adequate A (shared organizational culture) → schema-ward drift toward measurable proxies. The missing clarity constraint allowed each interpretive node to fill the gap with its local schema.

Case 2 — Micromanagement Drift

Leader intent: *"Deliver a reliable outcome."*

Management response:

- Frequent status reporting requirements
- Mandatory approval checkpoints
- Excessive process control layers
- Reduced operational autonomy

Employee recovered meaning: *"Compliance is more important than outcomes."*

The intended semantic content of the mission changed during amplification. As amplification pressure increased through control mechanisms, semantic recovery at the operational level inverted the original intent. What was meant to ensure reliability produced a system optimized for procedural conformance rather than outcomes.

Under the framework: high A (strong management presence), but the amplification system itself distorted rather than stabilized the signal — a pathological amplification failure. Low C (the original intent was not operationally specified) allowed the amplification system to substitute process compliance for outcome delivery.

Case 3 — Meeting Amplification Drift

Initial objective: *"Resolve database instability."*

After multiple coordination cycles:

- Reporting requirements expanded
- Documentation burden increased
- Additional stakeholders added
- Governance structures layered on

Recovered organizational meaning: *"Manage the process surrounding the database issue."*

The semantic center of gravity shifted from the technical problem to the coordination apparatus surrounding it. The original intent — resolution — was displaced by the amplification machinery — process management.

This corresponds structurally to the drift observed in highly vague texts under extended machine amplification: as iterations increased, the output began to reflect the amplification system's own structural tendencies rather than the original semantic content.

Synthesis — Human and Artificial Systems

3.1 Shared Drift Patterns

Across cultural, organizational, and artificial amplification systems, three consistent drift patterns emerge under the Meaning Theory framework:

- Structured content with strong anchoring preserves meaning under amplification.
- Vague or weakly specified content drifts toward the amplification system's own dominant schemas.
- Sufficient weakening of any single variable (S, A, or C) produces semantic instability regardless of the others.

In artificial systems, this produced the structured/vague gradient observed across all five models. In cultural systems, it produces the idiom/literal gradient documented across translation and cross-cultural pragmatics. In organizational systems, it produces the delegation/micromanagement drift patterns described above.

3.2 Differences Between Substrates

The mechanism of amplification differs between substrates:

- Machine systems: recursive re-encoding through model paraphrase, operating on token probability distributions
- Cultural systems: generational transmission through community membership and contextual reinforcement
- Organizational systems: active human interpretation filtered through local goals, schemas, and authority structures

These mechanistic differences may produce different drift signatures while reflecting the same underlying dynamic. This is consistent with the AI pilot finding that drift behavior was consistent within each model but varied between models — suggesting substrate-specific drift signatures within a shared structural framework.

3.3 The Substrate-Independence Hypothesis

If semantic stability failures emerge similarly across machine amplification systems, cultural transmission systems, and organizational coordination systems, then semantic stability may possess a partially substrate-independent structural layer.

Under this interpretation:

- Structural meaning governs semantic stability under amplification across substrates.
- Substrate-specific factors (model architecture, cultural schema, organizational power structure) shape the signature of drift without altering the underlying dynamic.

This is a hypothesis, not a demonstrated conclusion. The evidence presented here is illustrative and operational rather than statistically rigorous. A stronger test would require controlled experiments with comparable amplification pressure across human and artificial systems simultaneously.

Testable Predictions

If Meaning Theory contributes explanatory value beyond existing descriptive frameworks, it must generate predictions distinguishable from its source literatures.

Prediction 1 — Multiplicative Collapse

If semantic stability is multiplicative rather than additive, sufficiently weakening any single variable should produce semantic collapse regardless of the strength of the remaining variables. Additive models predict partial preservation proportional to surviving variables; Meaning Theory predicts collapse.

Prediction 2 — Differential Drift Signatures

Different forms of semantic drift should emerge depending on which variable is weakest: low C produces schema-ward reinterpretation; low S produces fidelity decay; low A produces opacity or untranslatability. This predicts not merely semantic loss but distinct failure signatures.

Prediction 3 — Chain Length and Drift Rate

Larger organizational chains should exhibit higher semantic drift rates. Micromanagement systems should display increased divergence between intended outcomes and recovered operational behavior. Organizations with strong shared context should exhibit greater semantic stability under amplification.

Prediction 4 — Cross-Substrate Drift Correspondence

Human cultural groups should display analogous group-specific semantic drift signatures under comparable amplification pressure to those observed across AI models. Drift behavior should be partially consistent within groups and variable between groups — as was observed across models in the AI pilot.

Scope and Limitations

This paper is exploratory and operational in nature. It does not prove Meaning Theory, validate the multiplicative form independently, establish universal semantic laws, or demonstrate substrate-independent cognition conclusively.

The semantic examples used are illustrative rather than mathematically rigorous measurements. No human-subject experiments were conducted. The organizational cases are constructed exemplars, not empirical observations from controlled studies.

The paper argues that:

- Semantic deviation across cultural and organizational systems is already well-documented across independent research traditions.
- These observations appear structurally consistent with the Meaning Theory framework.
- The framework generates testable predictions that distinguish it from purely descriptive prior work.

The framework remains hypothetical and requires further operational testing with blind raters, controlled amplification conditions, quantitative semantic similarity metrics, and larger sample sizes across both human and artificial systems.

Conclusion

Meaning Theory proposes that semantic stability is not a property of messages alone, but of the interaction between structural coherence, amplification systems, and clarity constraints.

This synthesis demonstrates that the dynamic observed in machine amplification systems — structured content surviving recursive paraphrase while vague content drifts — has structural analogues in both cultural and organizational human systems.

Idioms lose their meaning outside their originating cultural amplification system. Delegation chains transform intended outcomes into measurable proxies. Micromanagement systems invert the original semantic content of organizational missions. All of these correspond to predicted failure modes within the $M = S \times A \times C$ framework.

If this correspondence is genuine rather than coincidental, semantic stability may possess a partially substrate-independent structural layer relevant to communication design, organizational theory, and AI alignment research.

Meaning Theory does not seek to replace semiotics, pragmatics, cultural transmission theory, organizational communication research, or translation studies. It proposes that these fields may

be observing different manifestations of a shared semantic stability dynamic operating under varying amplification conditions across both biological and artificial systems.

Appendix — Cross-Cultural Semantic Stability Reference

The following table documents idiomatic expressions across language families and their corresponding semantic failure modes under Meaning Theory.

Expression	Literal Reading	Intended Reading	Failure Mode
"Break a leg" (EN)	Physical injury	Good luck	Low A outside culture
"Let's touch base" (EN)	Touching a base	Reconnect later	Contextual amplification dependency
"Circle back" (EN)	Move in a circle	Revisit topic later	Org-cultural dependency
"Chercher midi à 14h" (FR)	Look for noon at 2 PM	Overcomplicate something	Cultural semantic opacity
"Tomber dans les pommes" (FR)	Fall into apples	Faint	Low recoverability
"Kick the bucket" (EN)	Strike a bucket	Die	Idiomatic semantic collapse
"猫をかぶる" (JA)	Wear a cat	Pretend to be innocent	Cultural metaphor opacity
"空気を読む" (JA)	Read the air	Understand unspoken social context	Contextual-social amplification dependency
"Tomaten auf den Augen" (DE)	Tomatoes on one's eyes	Fail to notice the obvious	Metaphorical semantic remapping
"يد واحدة لا تصفق" (AR)	One hand does not clap	Cooperation requires multiple people	Collective-cultural reinforcement

Observation: across all examples, literal grammatical structure survives while intended semantic recovery depends on shared amplification systems. The persistence of this pattern across unrelated language families suggests that contextual amplification may function as a general stabilizing mechanism for semantic interpretation rather than a language-specific anomaly.

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